

Radio Environment Maps for 5G Cognitive Radio Network

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Abstract—Radio Environment Maps (REMs) have been introduced as a powerful tool in Cognitive Radio Networks to manage inter-transmitter interference. In the evolution of 5G heterogeneous ecosystem REMs are forced to be transformed to a more powerful tool hosting the diverse needs of an immense network. This paper reviews REMs and proposes an enhanced REM architecture for integration in the 5G network.

Keywords—REM, 5G, Cognitive Radio, Network Function Virtualization, SDN

I. INTRODUCTION

Cognitive Radio technology was introduced to answer the spectrum scarcity problem by utilizing the unoccupied spectrum in time, frequency and space [1] [2]. In Cognitive Radio Networks the unlicensed users also called and secondary users may occupy the spectrum if the licensed users also called primary users are not transmitting and if the interference they cause to the primary network is below a certain threshold. Secondary users can sense the spectrum for transmitting opportunities non-cooperatively or cooperatively. Many protocols have been introduced for CRNs for both scenarios with the cooperative schemes to achieve higher accuracy in most cases.

Radio Environment Maps concept was introduced to cooperatively collect, store and share information amongst the network users regarding the transmitter locations, spectrum usage maps, spectrum usage statistics in time and space, probabilistic models [3] and manage interference [4] [5]. Moreover REMs were introduced as more powerful tools to support CRN in its Cognitive Cycle of Radio Environment Awareness [6].

The immense 5G network would integrate a vast number of heterogeneous networks supporting an increasingly diverse set of services, applications and users from mobile users to IoT. In the meantime, on the quest for offloading the 5G users from spectrum management and thus reducing computation and energy consumption, REM arise as a collaborative tool residing in the network which efficient design would overcome the

overhead required for its update and increase overall network's performance.

II. THE REM CONCEPT

REM was first introduced by [6] as an integrated database that would store information locally or globally regarding spectrum usage, primary and secondary users transmission patterns, available channels, interference levels.

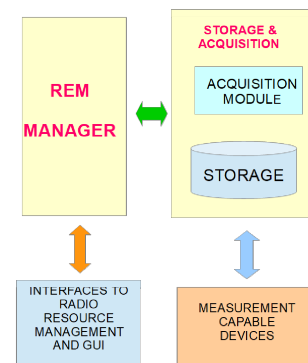


Fig. 1: The Basic REM Structure

The basic REM architecture consists of the REM storage and Acquisition module which is responsible of collecting the reports of the measurement capable devices which can reconfigure and storing the raw measurement data as well the output of the REM processes. The REM storage is a knowledge-based repository with spectrum occupancy information, the REM Manager is responsible for all the processing and REM generation and maintenance whilst the control application utilizes the REM information.

REM supports various CR applications such as i) hierarchical spectrum access, ii) spectrum sharing iii) intra-operator radio resource management iv) dedicated spectrum monitoring which is essential e.g. for intelligent transport

systems, self-organizing networks [7] v) interference management. On the other hand the information stored in the the REM storage and acquisition module can be classified as long term information which is updated slowly or short information which is updated more frequently [7].

The REM construction consists of exploiting the reports received by the Measurement Capable Devices and the estimations of the signal levels in the areas where there no MCDs. The accuracy of the REM depends on the i) resolution ii) the quality of data iii) the construction method iv) the REM data update period. If the information stored in the REM is outdated then the whole network's performance degrades.

The REM construction methods fall in three classes namely i) the direct methods (interpolation based) (Fig.2), ii) indirect (localization and propagation based) (Fig.3) iii) hybrid methods. In direct methods spatial interpolation is applied to the spatial RF- maps of the measured signals on a neighborhood basis. The most well-known direct methods are Inverse Distance Weighted (IDW), Nearest Neighbors (NN), splines, Natural Neighbors (NNI), modified Shepard's, Kriging, Gradient plus Inverse Distance Squared (GIDS). A survey on REM construction methods can be found in [7].

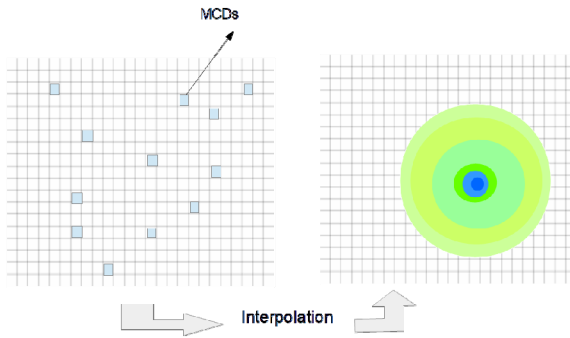


Fig.2: Direct RF-REM construction

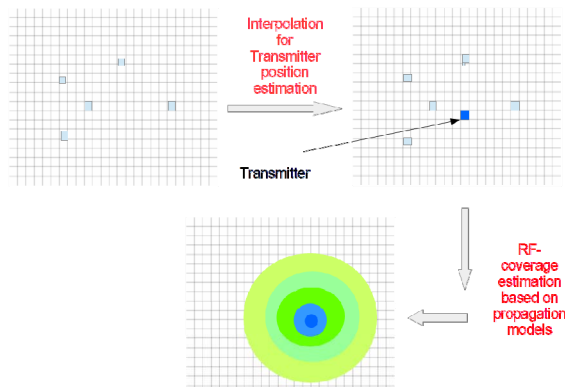


Fig.3: Indirect RF-REM construction

In indirect methods interpolation is applied to estimate the transmitter location and then the transmitter parameters and propagation models provided, the RF coverage at each pixel is estimated using the propagation model. There are two basic indirect methods i) LiVE and ii) SNR-aided method [8]. In LiVE method the received RF signal of the available MCDs are collected and then a propagation model is used to estimate the

transmitter power and location in a least square optimization problem. The propagation model that would be selected is essential for this method. In SNR-aided method the angle of arrival (AoA) are used and SNR data fusion and a propagation model to construct the REM around the transmitter [8]. In [9] an indirect self-tuning method was proposed. In [10] a hybrid approach combining transmitter localization and interpolation applied for urban LoS and NLoS environments. Transmitter localization methods have higher accuracy than interpolation methods in general. The authors in [11] propose a distributed Kalman filter to estimate position, velocity and power of transmitter. In [12] the Kriging interpolation method is improved and the model was evaluated in urban areas.

III. REM IN 5G

REM construction is a complex process integrating the transmitters' information, propagation models selection, interpolation methods selection, exploiting the available MCDs capabilities, mobility, the RF coverage maps, the cognitive radio spectrum management and in the same time perceiving the past experience into the Cognitive Radio Cycle of observing the environment, orienting, planning, designing, acting. Not only that but other issues have to be considered such as spectrum sharing policies, national and cross-border regulations, operator and cross-operator policies which govern the spectrum management process.

In the necessity of a layered REM model was exploited at [13] and at [9] in a heterogeneous LTE architecture of macrocells and femtocells based on the coverage maps. In [14] a study of application of REM in urban areas is presented addressing the issues that affect the RF-coverage in urban areas such as location, interference, fading, time.

In the vast heterogeneous 5G network the REM deployment should keep up-to-date the long and short term information in order to efficiently support the network and maintain the data rates, latency, QoE required levels of performance.

Thus, the following aspects arise:

- REM deployment close to network nodes to handle fast changing information, receive the MCDs reports, react to the local RF environment situations exploiting local powerful nodes or the RAN.
- REM deployment not close to network nodes to handle slowly changing information which can reside in the core network and the cloud.
- REM deployment to handle mobility
- REM deployment to handle cooperation between heterogeneous networks
- REM deployment to handle inter-operator functionality

The issues which will arise each time e.g. the intensive interpolation processing, transmitter localization, a change in the number of available MCDs, a change in the propagation model that should be used would be pass to the REM in the core network or the cloud or be handled locally on a periodic basis, on demand or as emergencies.

In [15] the authors propose a REM architecture aligned with the RAN virtualization concept which considers Network

Function Virtualization and an SDN- 5G network. Network Function Virtualization allows the network resources to be seen as pool of virtual resources that can be managed by physical or virtual operators. The SDN controller forces the decisions to the network. The proposed architecture mainly consists of the abstraction models such as interference maps which are used for short-term spectrum management, the REM and repository of regulations. The Spectrum Manager Application (SMA) will receive the information of the abstraction models which eventually will pass to REM and will be used on demand. The SMA will rely on different databases and the REM.

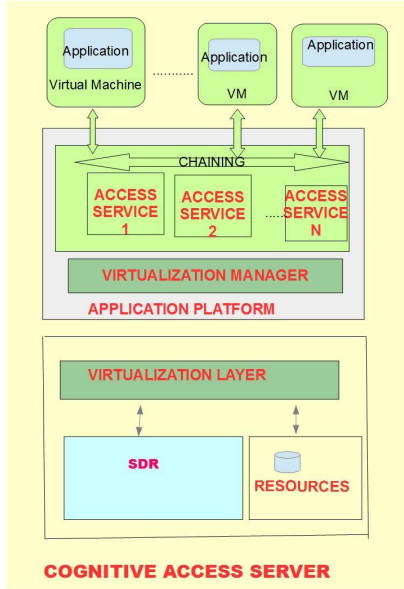


Fig.4: The Cognitive Access Server for the REM Deployment

Network Function Virtualization and SDN networks can be supportive in REM deployment. In [16] the cognitive radio functionality is offloaded at the Radio Access Network (RAN) or at the local powerful nodes in the form of access services provision for the Cognitive Radio Network by virtualizing the lower layer's functionality and resources and leveraging the connection to the core network and the Cloud connection. Lower layer Cognitive Radio Services and applications on the Edge Computing would increase capabilities not only within the RAN but within local mobile network on a peer-to-peer basis for access and backhauling. In the latter case, ad-hoc networks or vehicles would leverage powerful local nodes allowing them to be self-organized [16]. REM as a chained-service would benefit of the proposed architecture deployment as the MCDs would pass their reports to more powerful nodes and the REM deployment will be served by the proposed architecture in [16] (Fig.4 and Fig.5). However, as REM management encompasses national, regional and inter-operator policies and regulations, the necessity for new services arise for cross-operator, cross-border operability, QoE provision. For this reason an enhancement of the architecture [16] is proposed.

In order the REM deployment to respond to 5G challenges the provision of enhanced and abstract services on the top of the NFV chained services [16] arise. So, the NFV chained-services abstraction would be the basis for new abstract services composition. On the top of this layer would run an abstract service orchestration layer providing entries to the new services taking into account the policies, regulations, required level of performance and security issues. The abstract layers would allow operability between the heterogeneous networks, mobile nodes, cross-operators, QoS and QoE assurance as will make feasible the sharing and processing of virtual resources across the immense 5G network (Fig.6). Abstract service description scenarios would exploit the radio elements, radio scene, environment. The architecture would be enabled within the SDN context.

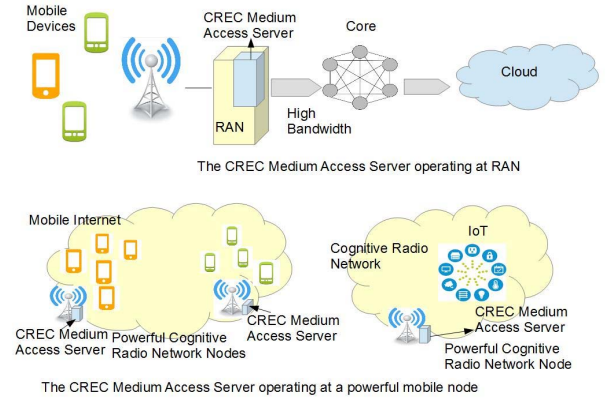


Fig.5: Cognitive Access Server deployment for the REM

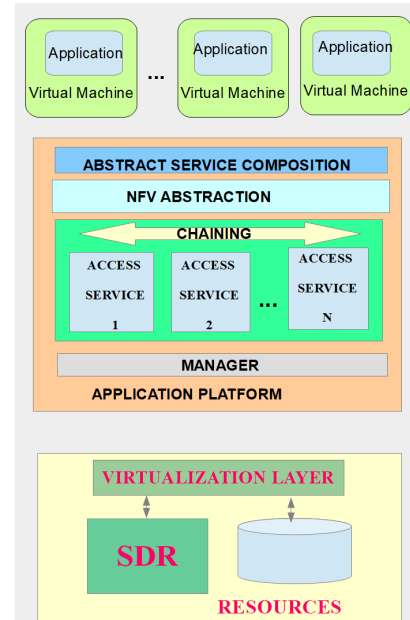


Fig.6: The Enhanced Cognitive Access Server for the REM deployment

V. CONCLUSION

The paper discusses the various issues of Radio Environment Maps construction and application and proposes the architecture [16] based on NFV and SDN network and an enhanced architecture of [16] for REM deployment in the 5G network taking into consideration the creation of new abstract services for cross-operator, cross-network operability and high level of QoE required in the immense and diverse 5G ecosystem.

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